

Pricing American Options with LSMC: Replication and Extension

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1 1. Financial context and relevance

American options can be exercised at any time up to maturity, so pricing them is an **optimal stopping** problem under the risk-neutral measure. For vanilla American puts, early exercise can be rational (especially deep in-the-money and with positive rates). For high-dimensional payoffs (basket options, path dependence, multifactor models), lattice/PDE methods become expensive or intractable, making simulation-based methods essential.

The 2001 Longstaff-Schwartz paper introduced a practical simulation approach (Least-Squares Monte Carlo, LSMC) that estimates continuation values via cross-sectional regressions and then applies backward dynamic programming. This became one of the most influential methods in buy-side/sell-side derivatives libraries.

2 2. Primary and extension papers

2.1 Primary paper

- Longstaff, F. A., & Schwartz, E. S. (2001). *Valuing American Options by Simulation: A Simple Least-Squares Approach*. The Review of Financial Studies.
- URL: <https://galton.uchicago.edu/~mykland/346W07/Longstaff.pdf>

2.2 Related/extended paper

- Tsitsiklis, J. N., & Van Roy, B. (2001). *Regression Methods for Pricing Complex American-Style Options*. IEEE Transactions on Neural Networks.
- URL: <https://www.mit.edu/~jnt/Papers/J086-01-bvr-options.pdf>

3 3. Methodology implemented

3.1 3.1 Longstaff-Schwartz (LSMC)

For each simulated path and time step (going backward):

1. Compute immediate exercise value (here for a put: $\max(K - S_t, 0)$).
2. Restrict to in-the-money paths.
3. Regress discounted realized future cashflows on basis functions of state (Laguerre basis in this implementation).
4. Estimated fitted value = continuation estimate.
5. Exercise if immediate payoff > continuation estimate; else continue.
6. Discount cashflows backward to $t = 0$.

3.2 Tsitsiklis-Van Roy style regression DP (extension)

This variant performs fitted value iteration with regression of discounted next-step value onto basis functions (using all paths). It is conceptually close to approximate dynamic programming and provides a useful comparison against classical in-the-money LSMC regression.

4 Data and experimental setup

- Underlying: AAPL
- Data source: Yahoo Finance (`yfinance`)
- Spot from latest close, volatility from annualized daily log-return std over 5 years
- Risk-free proxy: latest 13-week T-bill index ($\text{\textasciitilde}TRX$) / 100
- Maturity: 1 year
- Time grid: 50 time steps
- Monte Carlo paths: 100,000
- Strikes: $0.9S_0$, $1.0S_0$, $1.1S_0$

5 Python implementation and project structure

- `data/` : raw/derived data (`aapl_prices.csv`, `pricing_results.csv`)
- `src/` : implementation scripts (`lsmc.py`, `tsitsiklis_vanroy.py`, `data_loader.py`, `run_study.py`)
- `visuals/` : generated plots
- `report/` : this QMD report

6 Results

```
import pandas as pd
results = pd.read_csv('../data/pricing_results.csv')
results
```

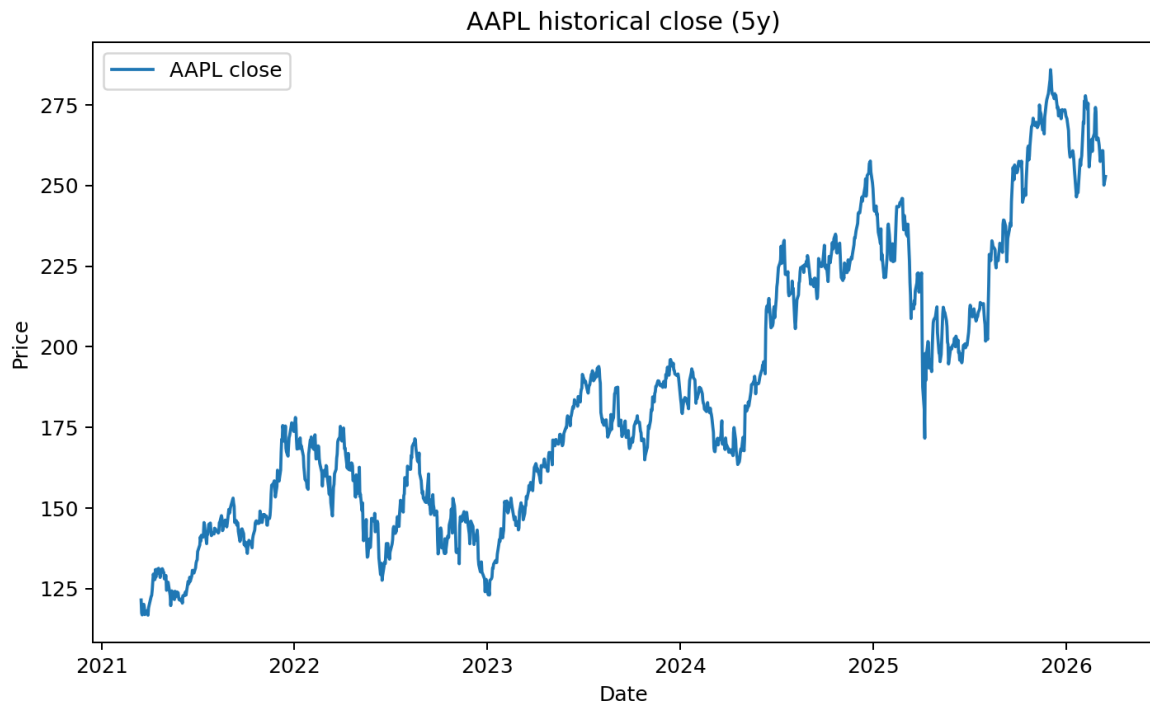
	ticker	spot	strike	maturity_years	volatility	risk_free	european_put_bs	lsmc_pric
0	AAPL	252.820007	227.538007	1.0	0.274248	0.03605	12.409216	12.712565
1	AAPL	252.820007	252.820007	1.0	0.274248	0.03605	22.842482	23.563214
2	AAPL	252.820007	278.102008	1.0	0.274248	0.03605	36.790064	38.313742

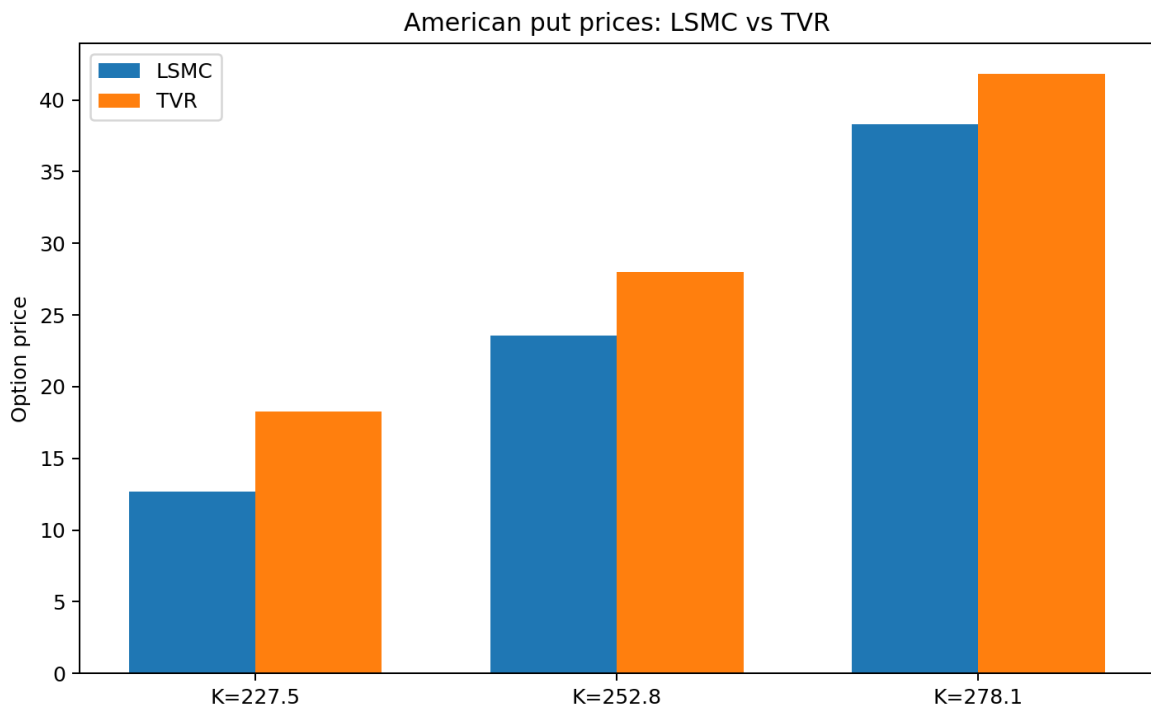
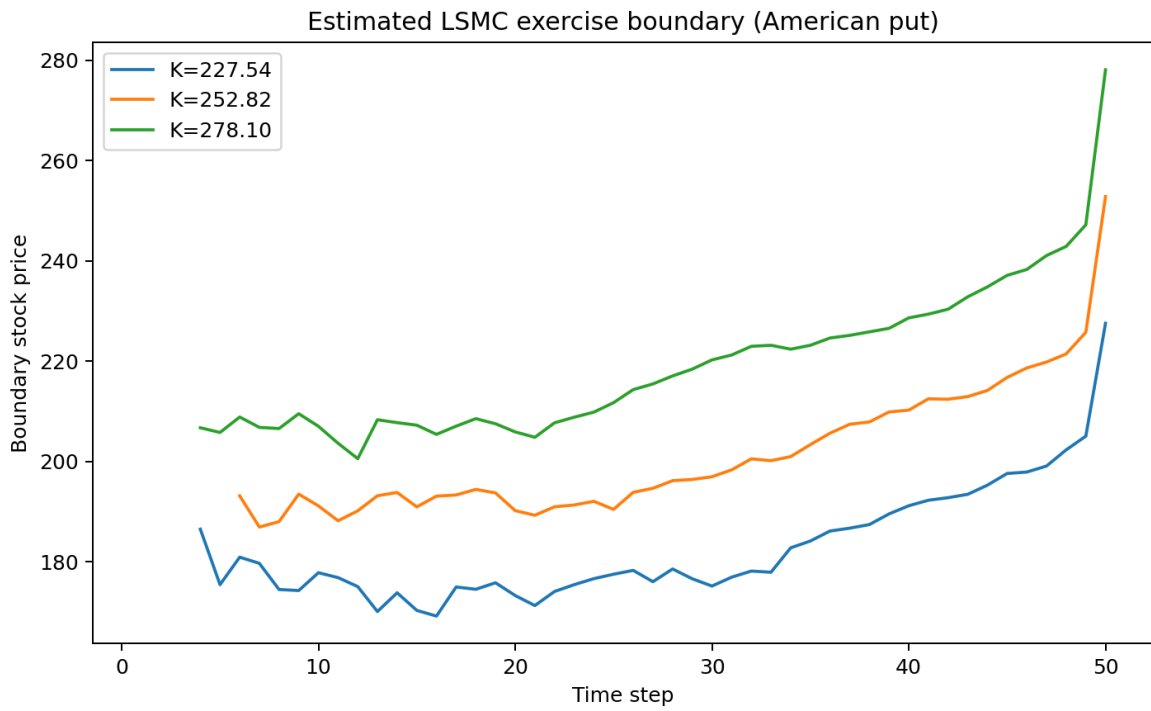
A quick consistency check is that American put values should be at least as large as corresponding European put values (same model inputs); this holds in the generated output.

```
(results[['strike','european_put_bs','lsmc_price','tvr_price']])
```

	strike	european_put_bs	lsmc_price	tvr_price
0	227.538007	12.409216	12.712565	18.257806
1	252.820007	22.842482	23.563214	27.985602
2	278.102008	36.790064	38.313742	41.849501

6.1 Visuals





7 7. Critical evaluation

7.1 Strengths of LSMC / regression simulation methods

- Scales better than trees/PDE in high dimensions.
- Flexible: supports path-dependent state variables and custom payoff logic.
- Natural fit for enterprise Monte Carlo risk stacks.
- Easy to parallelize.

7.2 Weaknesses / limitations

- **Basis risk / approximation error:** poor basis choice biases continuation estimates.
- **Look-ahead and finite-sample bias** risk if implementation is careless.
- **Exercise boundary instability** in sparse ITM regions.
- **Model risk:** under realistic markets, GBM with constant volatility/rates is often insufficient.
- **Tail events** underrepresented in standard MC unless variance reduction / stress design is used.

7.3 Does this address real-world needs appropriately?

Partly yes. LSMC is practically valuable and remains widely used. But institutions using it naively (simple GBM, fixed polynomial basis, weak diagnostics) may systematically misprice early-exercise features and risk exposures, especially in stressed regimes or products with complex state dependence.

7.4 Where institutions can run into trouble

- Underfitting continuation values for deep OTM/ITM states.
- Overfitting with high-order basis and limited effective sample.
- Unstable Greeks due to regression noise.
- Governance gaps: insufficient backtesting against benchmarks and market quotes.

7.5 Suggestions for improvement

1. **Richer dynamics:** local/stochastic volatility, stochastic rates, jumps when needed.
2. **Regularized / adaptive regression:** orthogonalized basis, cross-validation, or modern ML regressors with strong controls.
3. **Variance reduction:** antithetic/control variates, quasi-MC.

4. **Dual bounds** (e.g., Andersen-Broadie style) to quantify pricing bias intervals.
5. **Robust model validation**: benchmark against lattice/PDE in low dimensions, scenario stress testing, and out-of-sample stability checks.

8 8. Reproducibility

From project root:

```
python -m pip install yfinance pandas numpy matplotlib
python src/run_study.py
quarto render report/american_options_lsmc_study.qmd --to pdf
```

If PDF rendering tools are unavailable, render to HTML as fallback:

```
quarto render report/american_options_lsmc_study.qmd --to html
```